Contact

zihad.rafi16@gmail.com

www.linkedin.com/in/xihaad (LinkedIn)
xihaad.github.io/ahmed_rafi_hasan/ (Portfolio)
x.com/xihaad (Other)

Top Skills

Graph Neural Networks
Model Context Protocol (MCP)
AI Agents

Languages

English (Professional Working)
Bangla (Professional Working)

Certifications

Python
Intro to Deep Learning
SQL(Basic)
Programming Foundations with JavaScript, HTML and CSS
Intro to Machine Learning

Honors-Awards

Champion, Deep Learning Sprint, Fall 2022
1st Runner-Up, CSE Project Show, Summer 2022
7th in Dhaka Division
Cum Laude

Publications

A Dynamic Weighted Tabular Method for Convolutional Neural Networks
Accurate water level monitoring in Alternate Wetting and Drying rice cultivation using attention-based ConvNeXt architecture
A Novel Method for Imbalanced Data Classification based on Label Reassignment

Ahmed Rafi Hasan

Machine Learning Engineer | Deep RL Enthusiast | Kaggle Practitioner | Competitive Programmer
Dhaka, Bangladesh

Summary

I am a passionate and research-oriented Machine Learning Engineer with hands-on experience in multimodal machine learning, LLMs, and explainable AI solutions. Currently employed at Pivotly, I've contributed to business-driven GenAI systems and AI agents, while also engaging in academic research and publishing in reputed venues including IEEE Access and Computers and Electronics in Agriculture. My academic background includes a B.Sc. in CSE with a specialization in Data Science from UIU, where I earned Cum Laude distinction. I've held multiple roles, from instructor to ML engineer, and contributed to impactful projects in agriculture, healthcare, and scene understanding. My work bridges theoretical research with real-world applications, reflecting a strong foundation in machine learning, data handling, and system deployment. I also love problem-solving and geometry. My current research interest includes Cyber-Physical Systems, Deep Reinforcement Learning, Physics Guided Neural Networks, Embodied AI.Now-a-days, I am spending most of my time in learning Multi Agent Reinforcement Learning(MARL), Inductive Bias in Reinforcement Learning(IBRL), Control Theory, World Models.See details: https://xihaad.github.io/ahmed_rafi_hasan/

Experience

Pivotly
Machine Learning Engineer
August 2025 - September 2026 (1 year 2 months)
Still Water, MN, USA

Key Responsibilities & Contributions:

1. Pivotly Parse: Agentic Document Extraction System

*Designed and developed an agentic document extraction system for parsing complex architectural and technical drawings.

The impact of data locality on the performance of cluster-based under-sampling

Attention-Based Scene Graph Generation: A Review

*Tested LLM-based structured extraction pipelines using multi-turn and tool-augmented prompting strategies.
*Focused on improving structured JSON generation from LLMs for downstream engineering automation.
*Fine-tuned the Mask2Former model for image segmentation using custom-annotated architectural drawing datasets and built document extraction prototype.
*Conducted research on Graph Neural Networks (GNNs) for structured understanding of architectural drawings.

2. Multi-Agent Conversational Recommendation Systems (CRS)
*Designed intelligent agent-based systems that collaborated to deliver personalized recommendations and dynamic decision support.
*Conducted R&D on agentic OCR systems, multimodal models, and state-of-the-art open-source architectures.

3. AI-Powered Search Solutions for Client Projects
*Built domain-specific AI search experiences using retrieval-augmented generation (RAG), natural language querying, and custom ranking pipelines.

4. AI Feature Development for the Pivotly Platform
*Contributed to product-level AI functionality(Natural Language to cosmosDB query generations), including workflow automation and LLM integration.
*Developed MCP-based integrations prototypes and context-aware interactions.

5. In-House R&D and Prototype Development
*Worked on first AI feature research and proposed AI feature development list based on market demands, AI capabilities and Pivotly's business goals.
*Worked on internal initiatives involving LLM orchestration, prompt engineering frameworks, and document intelligence to accelerate AI adoption across the ecosystem.

Silicon Orchard Ltd.
1 year 6 months

Machine Learning Engineer
February 2025 - August 2025 (7 months)
Dhaka, Bangladesh

My projects cover the following areas at SOL:
1. Physics guided/informed/encoded Neural Network(PGNN, PINN, PENN)

2. Azure API Management
3. LLM Agent, RL
4. Multi Agent Conversational Recommendation System
5. Visual Search

Developed and deployed a full Machine Learning pipeline for the product: SOLCHAT. Have experience on API integration and ML project deployment. Entry-level understanding of cloud platforms, with primary knowledge of Microsoft Azure services, focusing on deployment, resource management, and basic service configurations.

Jr Machine Learning Engineer
March 2024 - February 2025 (1 year)
Dhaka, Bangladesh

Areas of Experience:
1. LLM and Explainable AI
2. Multimodal Machine Learning
3. Class Incremental Learning
4. Online Learning
5. Computer Vision
6. Distributed Training

Nodes Digital Limited
1 year 2 months

Jr Machine Learning Engineer
September 2023 - February 2024 (6 months)
Dhaka, Bangladesh

1. Project title: " Feasibility study of IoT-based Precision Agriculture for sustainable crop production in Bangladesh". This project is being developed by both NDL and BARC(Bangladesh Agricultural Research Council) and funded by KGF(Krishi Gobeshona Foundation).
2. Project title: "Alternate wetting and drying (AWD)". Farmers have largely not adopted the AWD practice in the production of rice. The goal of this project is to create, develop, and test a prototype IoT-based smart AWD system for irrigation-based agriculture. IoT-based AWD system enables fair water distribution among the farmers as well as ongoing monitoring and real-time decision-making.

My role is to design & develop the Machine learning algorithm to empower IoT tools to be smart.

Machine Learning Intern
January 2023 - August 2023 (8 months)
Dhaka, Bangladesh

United International University
1 year 7 months

Parttime Lecturer/ Instructor
May 2023 - September 2023 (5 months)
Dhaka, Bangladesh

Course Teacher of Digital Logic Design and Discrete Mathematics.

Undergraduate Teaching Assistant
March 2022 - May 2023 (1 year 3 months)
Dhaka, Bangladesh

I have been selected as an "Undergraduate Teaching Assistant" for the following courses.

Artificial Intelligence Laboratory (Spring 2022), Data Structure & Algorithm I Laboratory (Spring 2022), Data Mining (Summer 2022), Data Structure & Algorithm II Laboratory (Fall 2022), Structured Programming Language Lab (Spring 2023).

UIU Computer Club
Programming Wing Executive
September 2021 - May 2022 (9 months)
Dhaka, Bangladesh

Education

United International University
Bachelor of Science - BS, Computer Science · (2018 - 2023)

Notre Dame College
Higher Secondary Certificate, Science · (2015 - 2017)

B.A.F Shaheen College (Kurmitola)
Secondary School Certificate · (2004 - 2014)